Harmonic Conjugate:

1) Find the harmonic conjugate of $u(x,y) = x^3 - 3xy^2 - 5y$.

Solution: $u(x,y) = x^3 - 3xy^2 - 5y$.

$u_x = v_y$, $u_y = -v_x$.

$v_y = 3x^2 - 3y^2$, $v_x = 6xy + 5$.

Integrating,

$v(x,y) = 3x^2y - y^3 + c$, $v = 3x^2y + 5x + c$.

$\therefore v(x,y) = 3x^2y - y^3 + 5x + c$.

2) Verify that the given function u is harmonic and find harmonic conjugate v.

Solution: i) $u = xy + x + 2y$.

$u_x = y + 1$. $u_y = x + 2$.

$u_{xx} = 0$, $u_{yy} = 0$.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

u is harmonic.

C-R equations:

$u_x = v_y$, $u_y = -v_x$.

$v_y = y+1$ $\qquad$ $v_x = -x-2$.

Integrating,

$v = \frac{y^2}{2} + y + c$. $\qquad$ $v = -\frac{x^2}{2} - 2x + c$.

$\therefore v(x,y) = \frac{y^2}{2} + y - \frac{x^2}{2} - 2x + c$.

(ii) $u(x,y) = 4xy^3 - 4x^3y + x$.

3) Find the constant 'a' so that $u = ax^2 - y^2 + xy$ is harmonic.

Solution:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$$2a - 2 = 0. \Rightarrow \boxed{a = 1}$$

4) Find the value 'm' of function $u = 2x + x^2 - my^2$ is harmonic.

Solution:

$$u_{xx} + u_{yy} = 0.$$

$$2 - 2m = 0.$$

$$\Rightarrow \boxed{m = 1}.$$